Karnan Thamilchelvan

k3thamil@uwaterloo.ca | 647-548-6236 | Website | LinkedIn

SKILLS

Hardware/PCB Design
PCB Design, Soldering & PCB Rework, Circuit Debugging, EasyEDA, KiCAD, SolidWorks

Embedded/Digital Systems
SystemVerilog, VHDL, Quartus (FPGA), STM32Cube, Arduino IDE, Microcontrollers, LTSpice

Software/Languages
C++, C, Java, Python, MATLAB, SQL, Flutter, Jira, VSCode

Lab Equipment
Oscilloscope, Function Generator, Multimeters

EXPERIENCE

Robotics & Controls Designer, *Jet Automation* — Jan 2026 – Apr 2026 | Mississauga, ON

- Engineered an automated packaging system using an Omron TM12 collaborative robot; programmed precise robotic motion paths in TMFlow to optimize cycle times (10-15s) across 4 unique box configurations.
- Integrated Omron Sysmac PLCs with industrial I/O, vacuum end-effectors, and 4 digital sensors to build a reliable, sensor-driven automation sequence and production timeline.
- Designed and deployed a custom HMI in Sysmac Studio for box-size selection, automation controls, industrial I/O monitoring, and emergency-stop integration through PLC mapping.
- Executed Factory Acceptance Testing (FAT), systematically troubleshooting firmware anomalies and validating hardware integrity for production-ready robotic nodes.

Hardware & Embedded Systems Intern, *University of Waterloo - IDEAs Clinic* — May 2025 – Aug 2025 | Waterloo, ON

- Developed a Mechatronic Resonance System from the ground up utilizing an Arduino UNO R4 Wi-Fi, proximity/motion sensors, and a custom embedded control unit.
- Owned the full PCB lifecycle by capturing schematics in EasyEDA, routing layouts, and hand-soldering/fabricating 50+ production boards for lab integration.
- Modeled in SolidWorks and fabricated 220+ 3D-printed and machined parts, assembling 40 complete units actively utilized in the ECE core curriculum.
- Authored and delivered FPGA programs in Quartus to introduce intermediate digital design, state machines, and embedded hardware concepts to undergraduate students.

Hardware Engineer, *Ascenix (Formerly Fallyx)* — Sep 2024 – Dec 2024 | Toronto, ON

- Designed custom IoT PCBs integrating ESP32 microcontrollers and barometric pressure sensors for a medical-assistive fall detection device deployed across 50+ senior care facilities.
- Optimized system reliability by calibrating 6-axis Inertial Measurement Units (IMUs) and refining embedded digital signal processing algorithms to increase fall-detection accuracy.
- Prototyped form-fitting enclosures in SolidWorks, utilizing 3D-printing technologies to deliver a durable, ergonomic, and production-ready physical device.

PROJECTS

LANA Vision (Capstone Project), *Python, C, MCUs, Hardware Architecture* — May 2026 – Apr 2027

- Co-designing an autonomous guide robot by architecting an end-to-end communication pipeline that bridges an onboard server with a cloud-hosted Agentic AI model to enable mapless, natural-language-guided indoor navigation.
- Developing low-latency firmware for the hardware controller and writing custom driver interfaces to process real-time streaming data from camera/sensor arrays and translate high-level AI model decisions into physical RC car actuation.

Smart-Irrigation Controller, *STM32, STM32CubeIDE, C, KiCad*

- Developed an automated irrigation system for terraced landscapes, integrating PWM-controlled pumps, servo-driven water distribution, and ultrasonic sensors to optimize water use based on real-time reservoir levels.
- Designed a custom PCB with an STM32 microcontroller and timer board for precise motor control and monitoring.

EDUCATION

University of Waterloo, — Sep 2022 - Apr 2027 | Waterloo, ON

Candidate for Bachelor of Applied Science – BASc, Computer Engineering

Relevant Courses: Digital Hardware Systems, Algorithms and Data Structures, Compilers